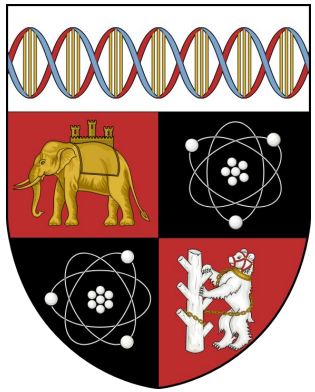
Feistel Tools: Reprogramming and Query-Recording for QRPs

Yu-Hsuan Huang, Andreas Hülsing,
Varun Maram, Silvia Ritsch, Abishanka Saha





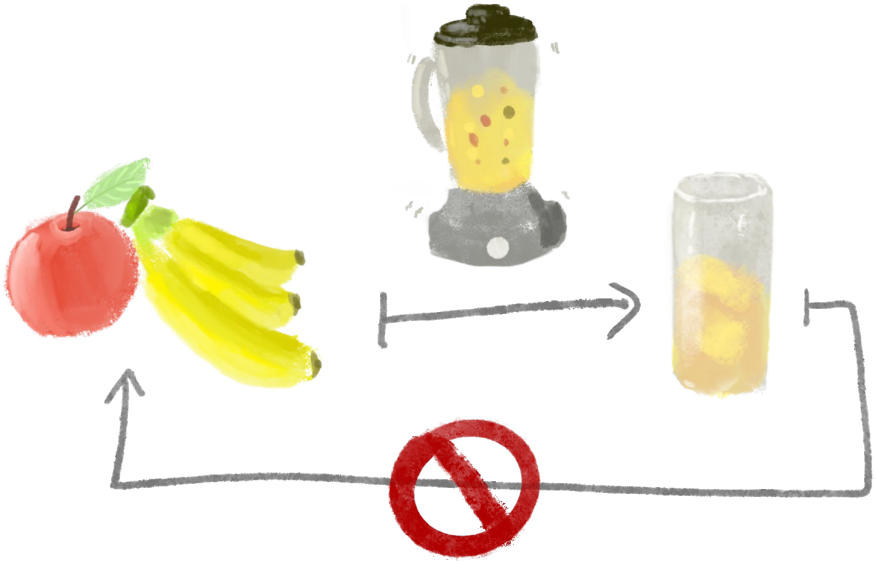




Illustrations/slides due to myself, ChatGPT, online meme generator, and Silvia

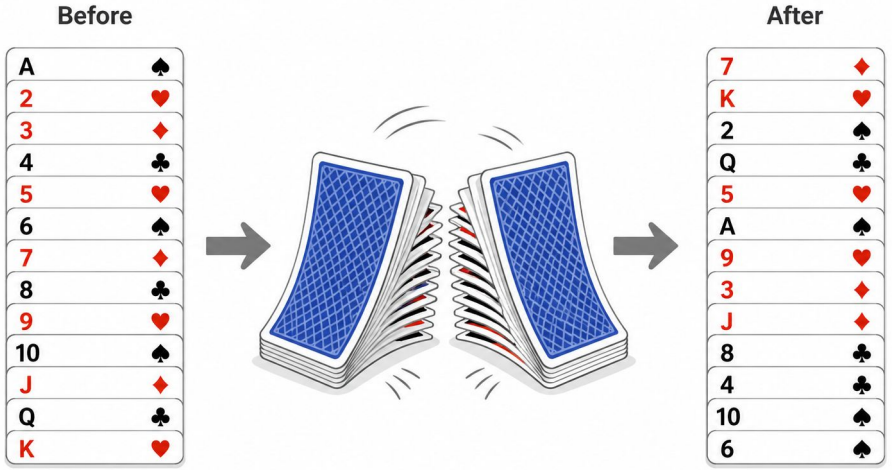
Symmetric Primitives...

One-Way Functions:
Easy to evaluate, hard to invert.



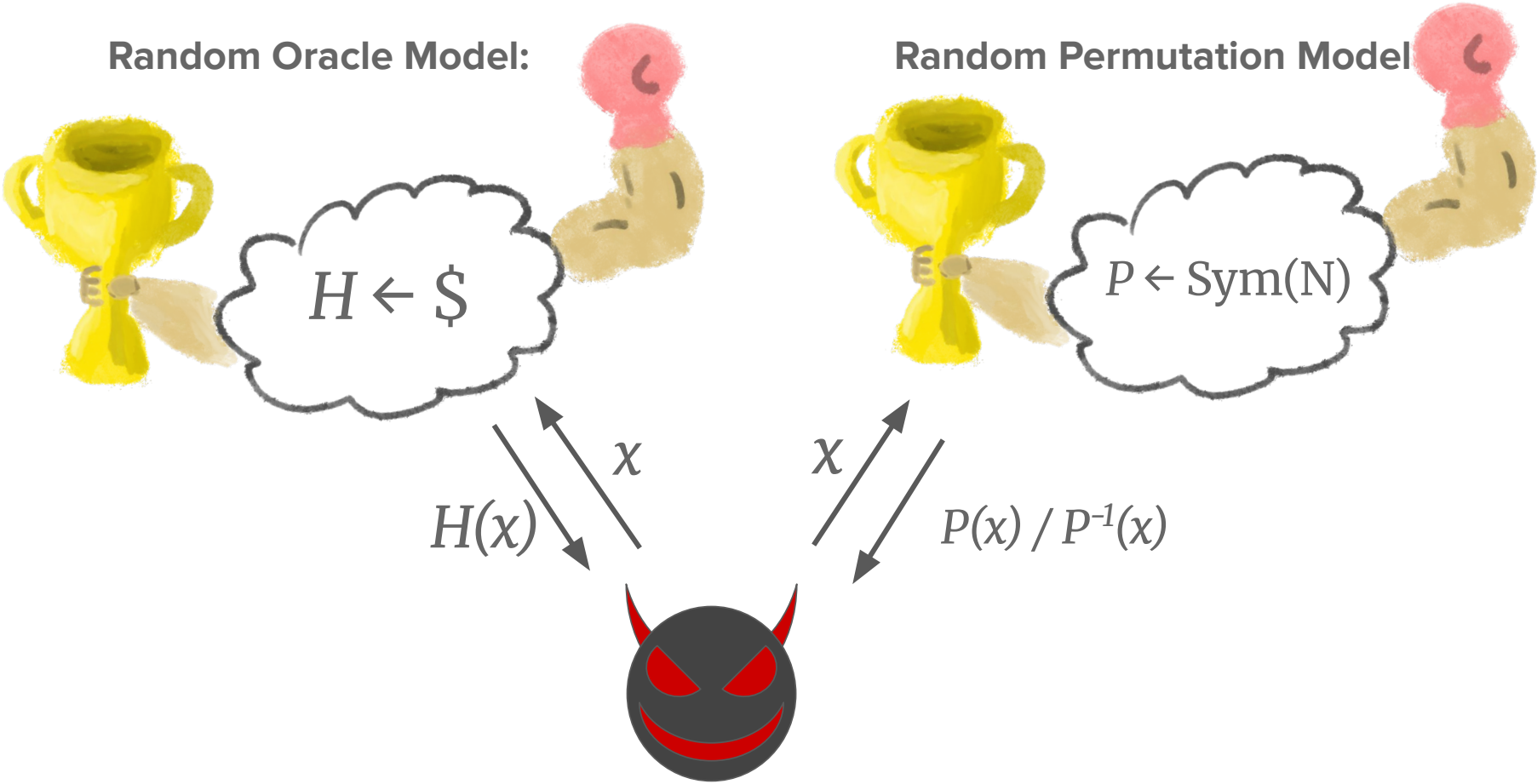
hash function, compression function ...

Permutations:
Easy to evaluate, often easy to invert.

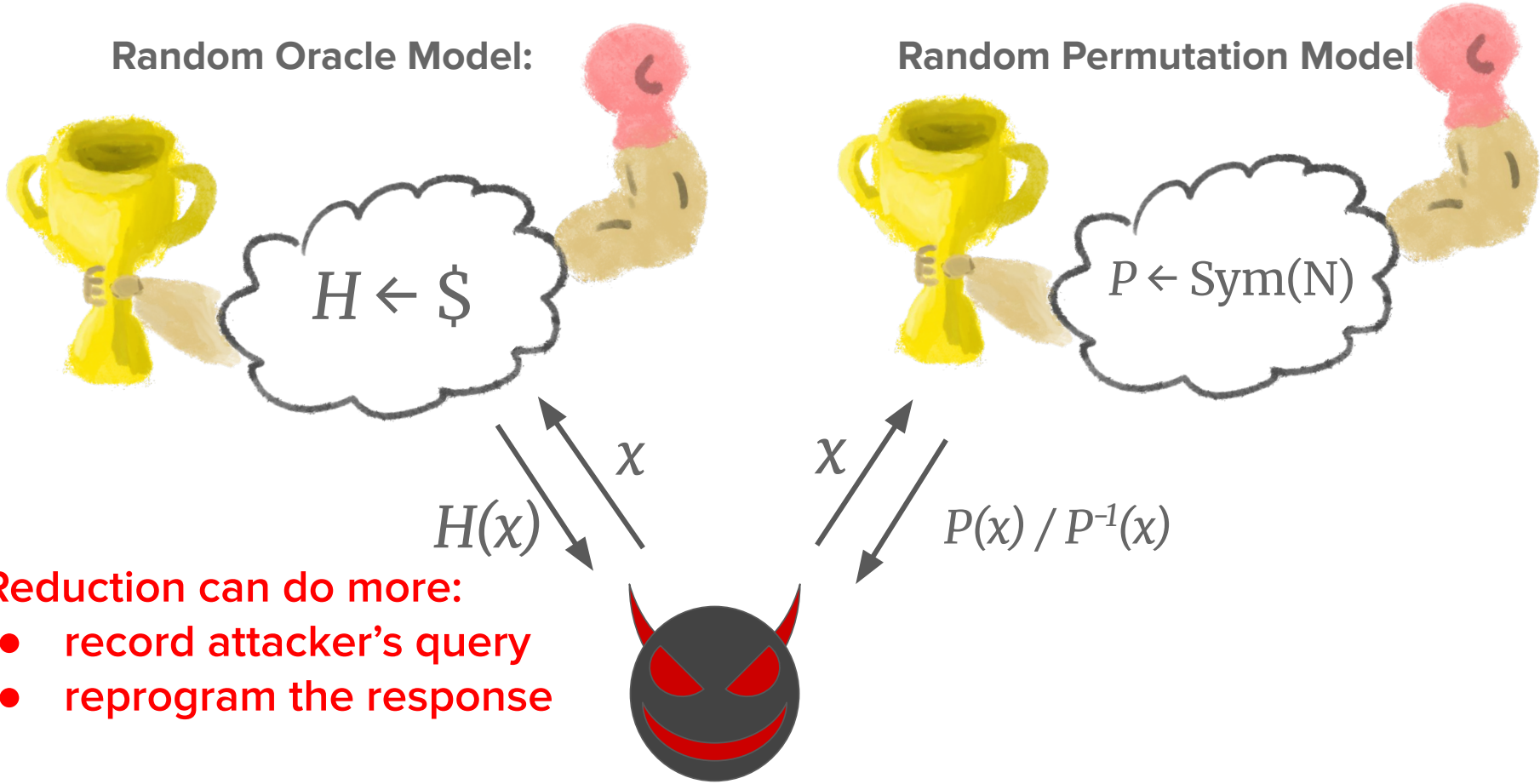


(fixed key) block cipher, round of sponge ...

Symmetric Primitives...



Symmetric Primitives...



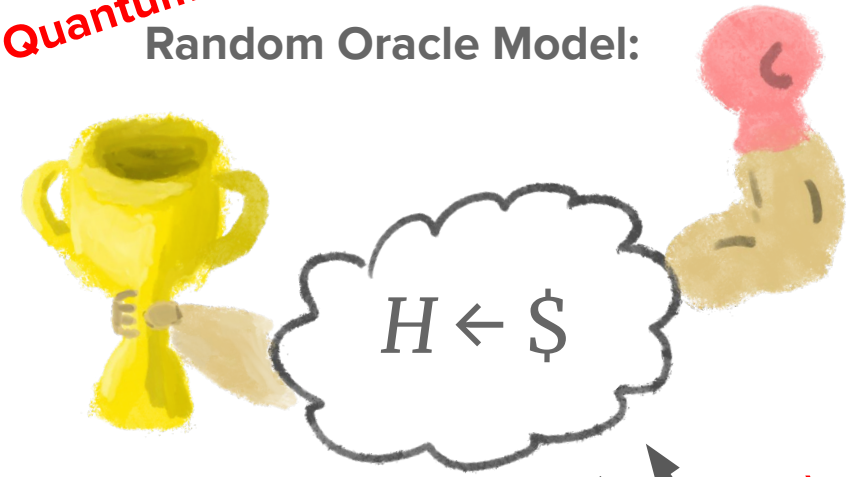
Reduction can do more:

- record attacker's query
- reprogram the response

Symmetric Primitives...

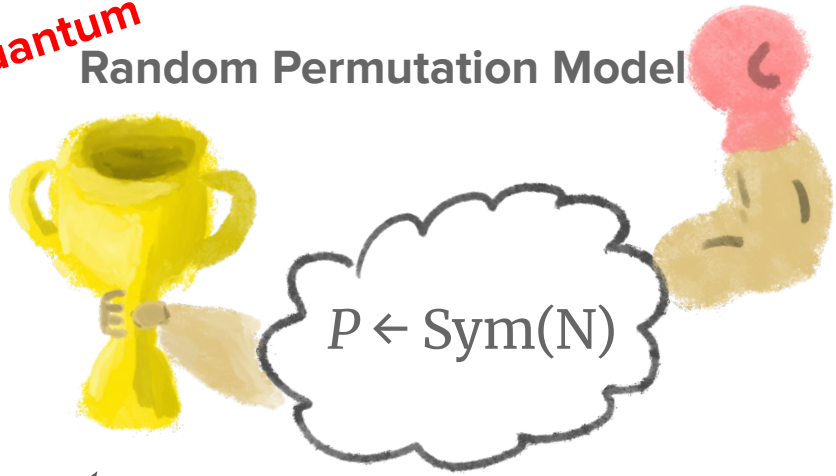
Quantum

Random Oracle Model:



Quantum

Random Permutation Model



$|x\rangle$ $|x\rangle$

$|H(x)\rangle$ $|P(x)\rangle$ / $|P^{-1}(x)\rangle$

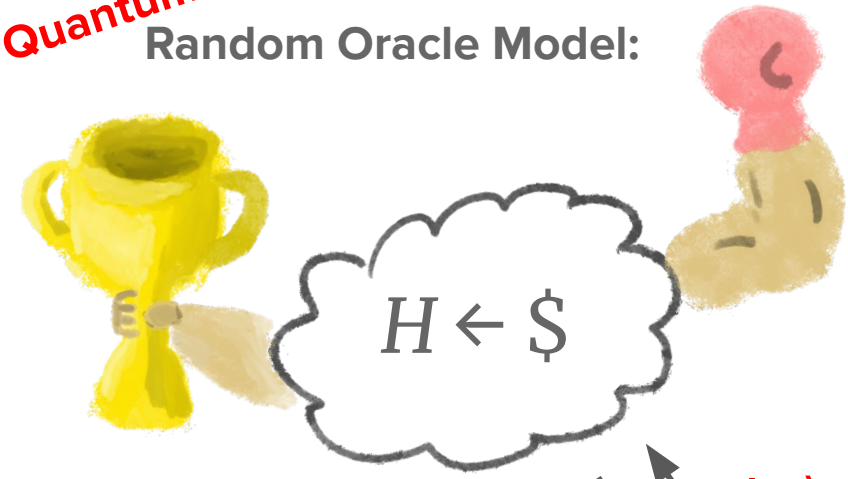


- record attacker's query?
- reprogram the response?

Symmetric Primitives...

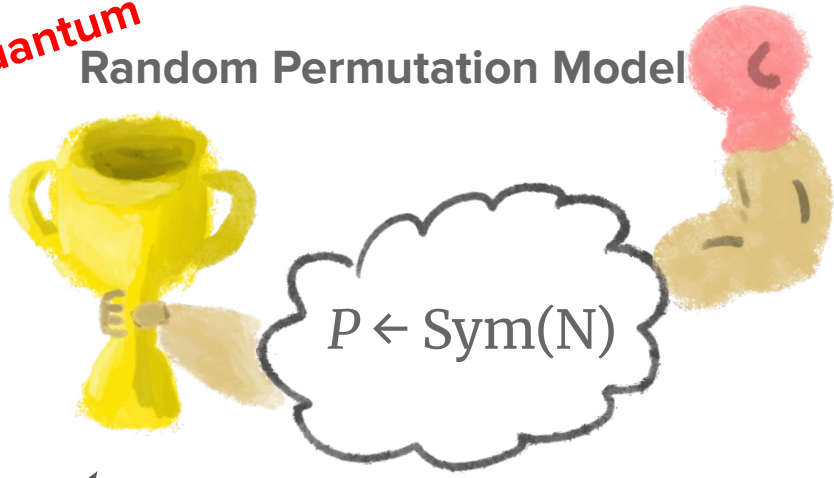
Quantum

Random Oracle Model:



Quantum

Random Permutation Model



$|x\rangle$ $|x\rangle$

$|H(x)\rangle$ $|P(x)\rangle$ / $|P^{-1}(x)\rangle$

No cloning theorem?

- record attacker's query?
- reprogram the response?

superposition response?



A QROM Toolbox (Prior)

History-free reductions

Introduction of QROM [BDFLSZ11]

Small-range distribution [Zhandry12]

Query Recording/Extraction

Compressed oracle [Zhandry19]

Online extraction [DFMS21]

Reprogramming "useful" queries

Measure-and-reprogram (M&R) [DFMS19]

Measure-and-reprogram 2.0 [DFM20]

Coherent measure-and-reprogram [CGLS24]

Oracle-dependent advice

Multi-instance game [CGLQ20]

Alternating measurement [Liu23]

Reprogramming hidden points

One-way to hiding (O2H) family: [AHU19, BHHHP19, KSSSS20, GLX24, LGXZ26]

Reprogramming unpredictable points

Adaptive O2H [Unruh17]

Adaptive reprogramming [GHHM21]

Pan-Zeng reprogramming [PZ24]

Classical-to-quantum lifting theorems

Lifting hardness of search games [YZ21]

Lifting PRG constructions [KS24]

A QRPM Toolbox (Prior)

Query Recording

Limited to single input-output [MMW25]

Reprogramming hidden points

One-way to hiding (O2H) family: [AHU19, BHHHP19, KSSSS20, GLX24, LGXZ26]

Reprogramming unpredictable points

Resampling lemmata [ABKM21, ABKMS24, Hos25]

Classical-to-quantum lifting theorems

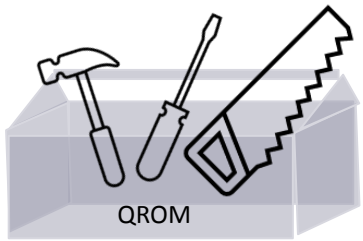
Lifting hardness of search games [CHLYY25]

The Gap in Prior Toolboxes

Quantum Random Oracle Model:

Rich QROM toolbox:

- Query recording
- many flavours of reprogramming:
  - measure and reprogram
  - one-way to hiding
  - adaptive reprogramming
- Techniques for auxiliary inputs
- many more...

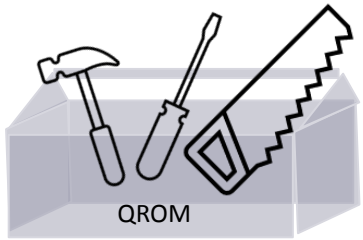


The Gap in Prior Toolboxes

Quantum Random Oracle Model:

Rich QROM toolbox:

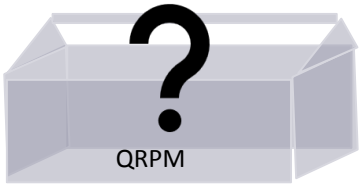
- Query recording
- many flavours of reprogramming:
  - measure and reprogram
  - one-way to hiding
  - adaptive reprogramming
- Techniques for auxiliary inputs
- many more...



Quantum Random Permutation Model:

Relatively fewer was known for QRPM:

- Very restricted query recording [MMW24]
- Fewer reprogramming tools proven via tailored approach [ABKM21, CHLYY25]
- Almost no auxiliary-input techniques



The Gap in Prior Toolboxes

Quantum Random Oracle Model:

Rich QROM to
- Query record
  - ...
  - ...
  - adaptive repro
- Techniques for au... inputs
- many more...

Quantum R...om Permutation Model:

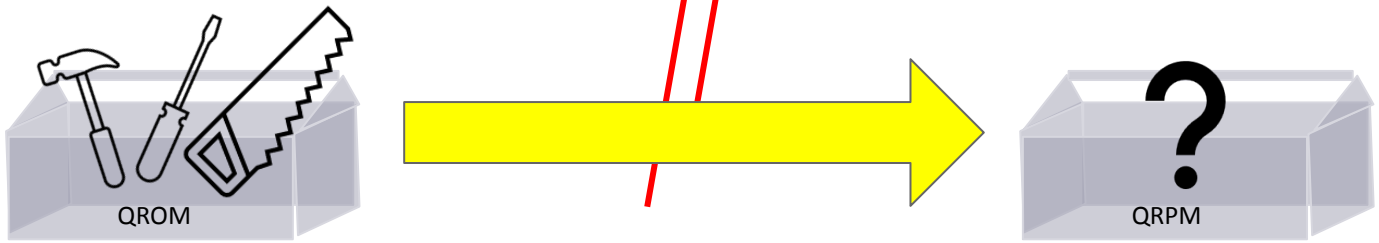
M:

...[MMW24]

...Y25]

...niques

This work: a framework lifting QROM tools → QRPM tools!



Our Contribution

We introduce *P3FQ*: a perfect*, efficient QRP simulator that recovers QROM techniques:

1. Query recording
2. Adaptive reprogramming (even with advice!)
3. Measure-and-reprogram

As applications:

1. Recovering query complexity bounds and obtaining new ones.
2. A security proof of Duplex-Sponge Fiat-Shamir (DSFS) in QRPM.

*up to an arbitrarily small error

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Concurent: A simpler simulator with simulation loss [Carolan25]

Subsequent works:

1. A perfect but possibly inefficient simulator [FLMNW26, GY25]
2. Improved simulation loss [CM26, Kim26]
3. Generalization to ideal-cipher [BH26, CM26]
4. QROM Reprogramming with quantum advice [HL26]

*up to an arbitrarily small error

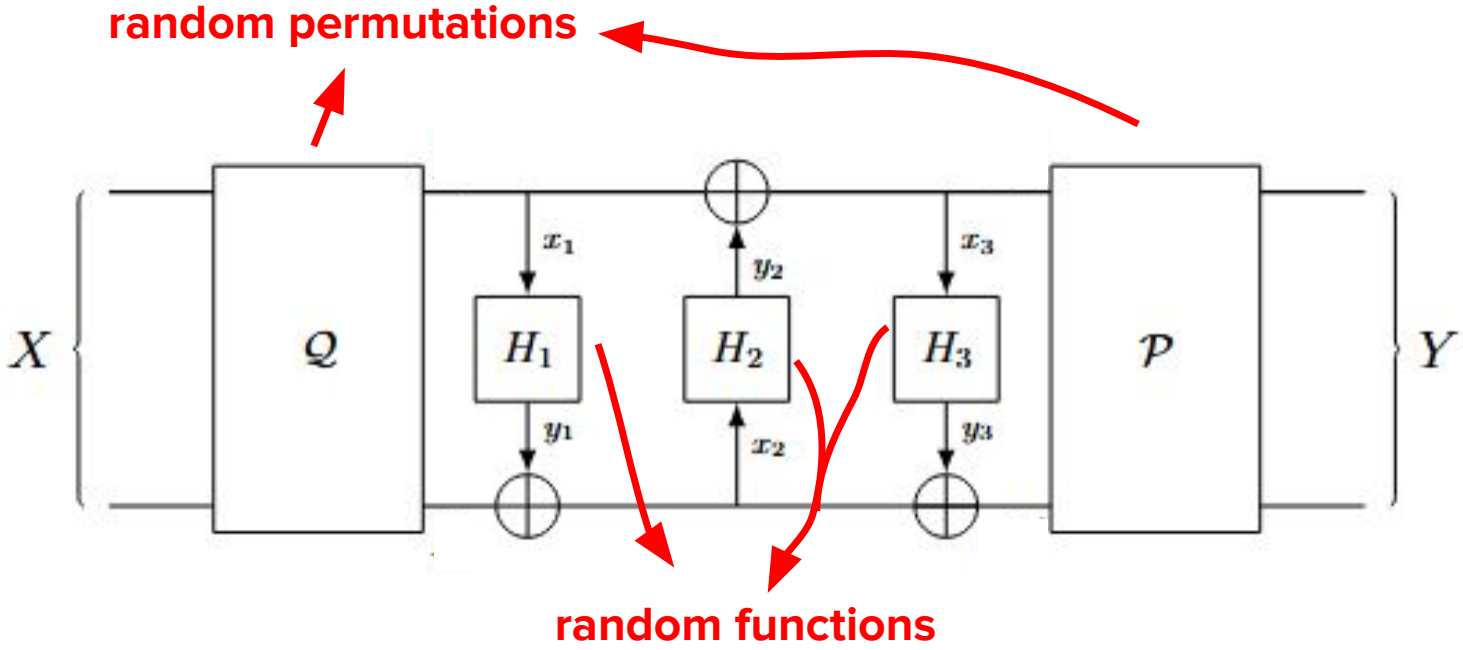
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Core Idea

P3FQ, a random permutation simulator:

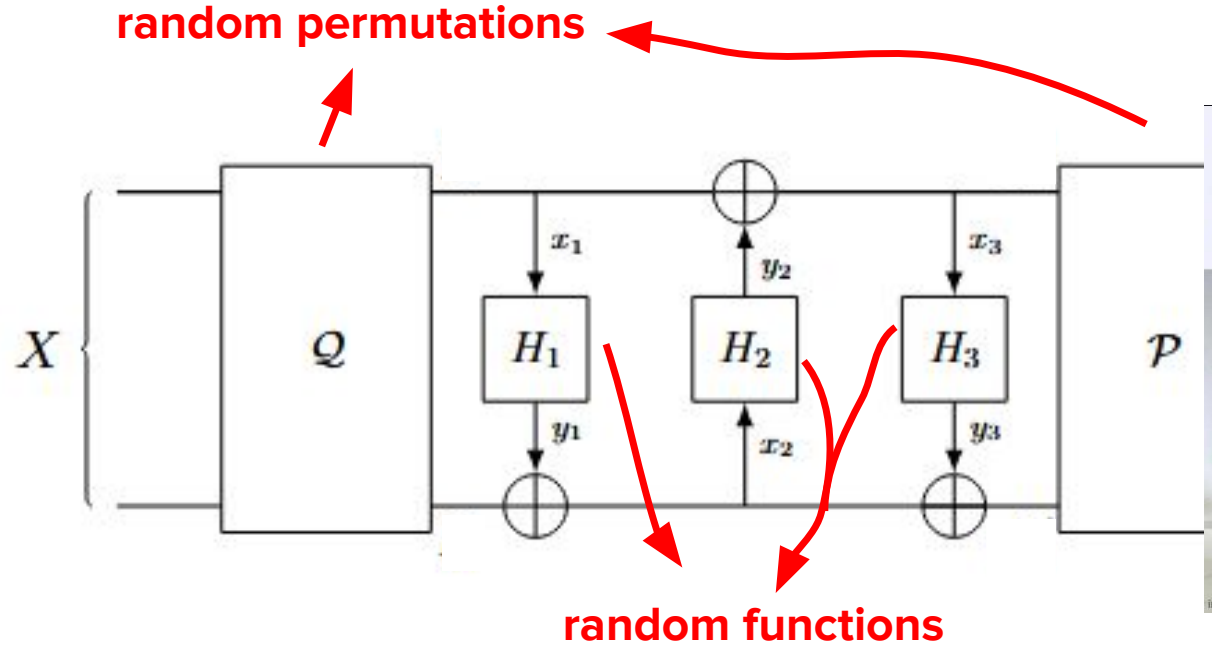
$$Y = P \circ [3\text{-round Feistel}] \circ Q(X)$$



Core Idea

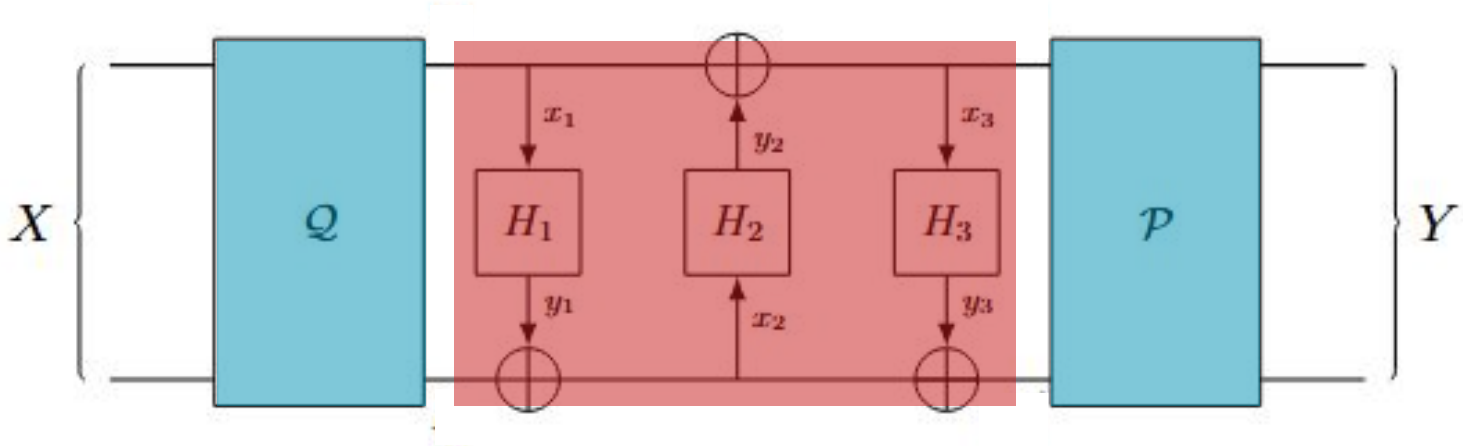
P3FQ, a random permutation simulator:

$$Y = P \circ [\textit{3-round Feistel}] \circ Q(X)$$





Core Idea



Outer random permutations:

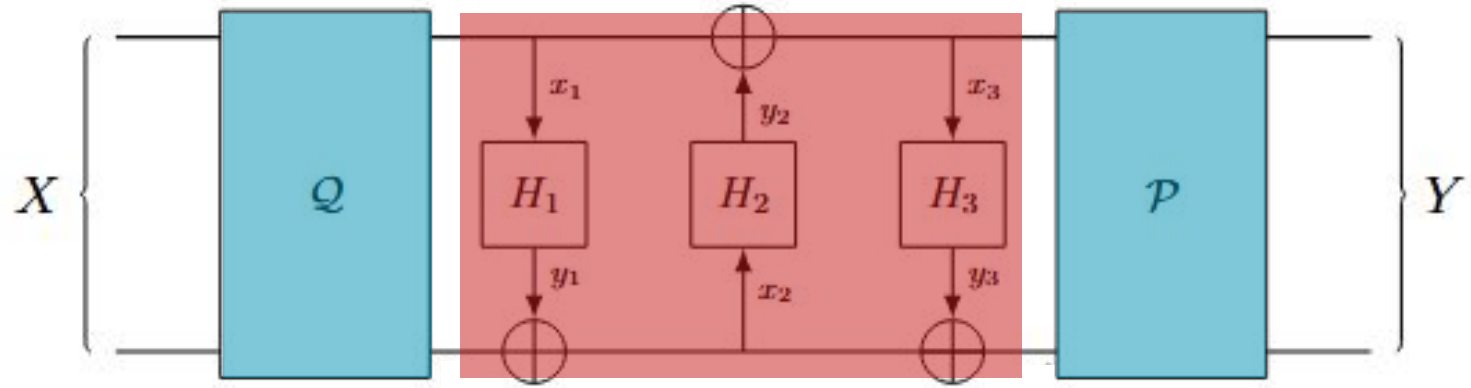
- **provide indistinguishability**
- **hide internal Feistel, and hide each other.**

Inner random oracles import QROM tools:

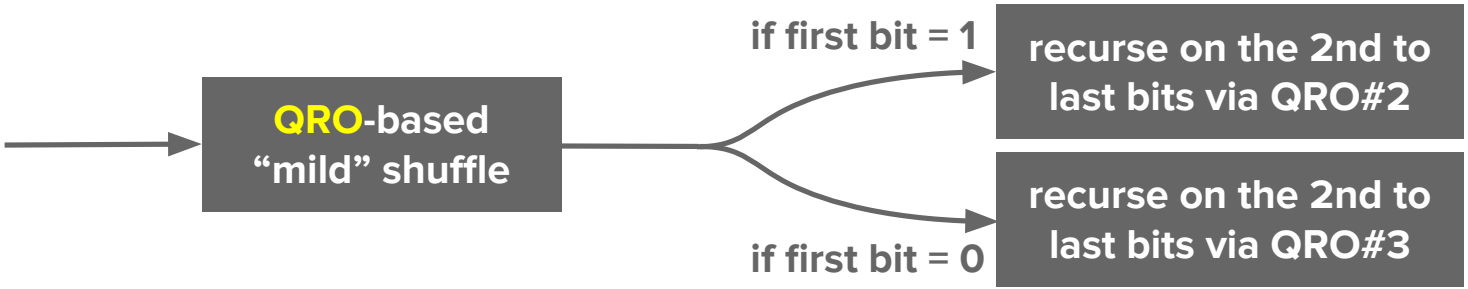
- **reprogramming**
- **query recording, etc.**

Core Idea

Also: ∃ efficient (stateful) simulation

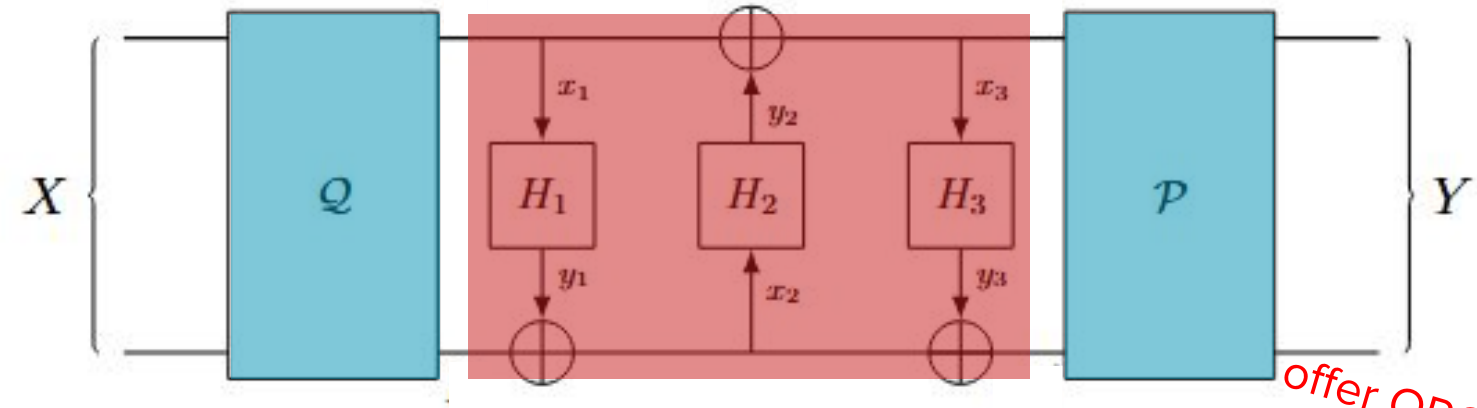


1. **QRO simulation** via Zhandry's compressed oracle technique
2. **QRP simulation** via the "mix-and-cut" technique:



Core Idea

Also: ∃ efficient (stateful) simulation



offer QROM tools

1. **QRO simulation** via Zhandry's compressed oracle technique
2. **QRP simulation** via the "mix-and-cut" technique:

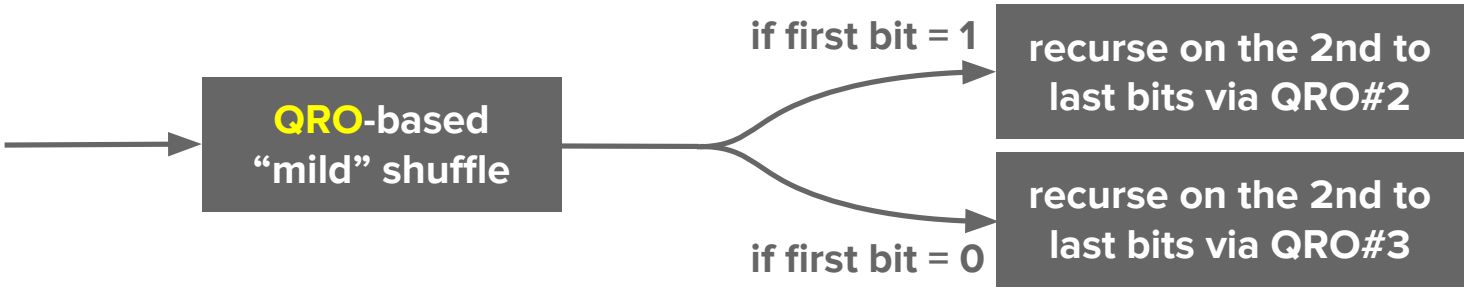
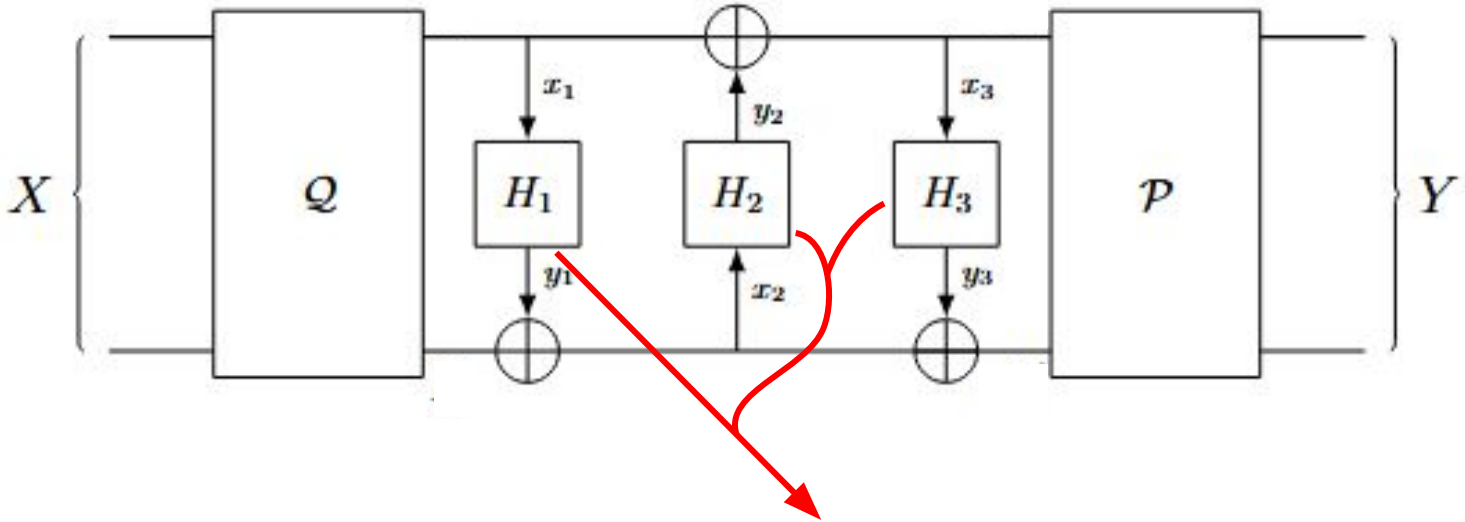


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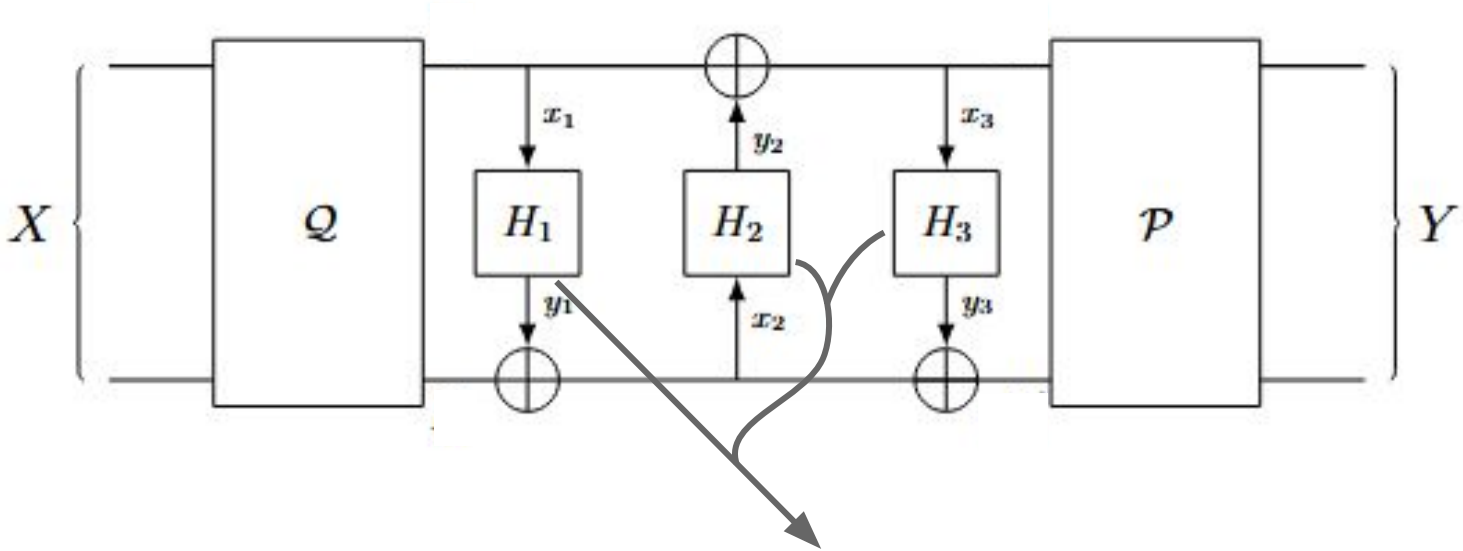
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Lifting Query Recording



Compressed oracle: record H_1, H_2, H_3 queries

Lifting Query Recording

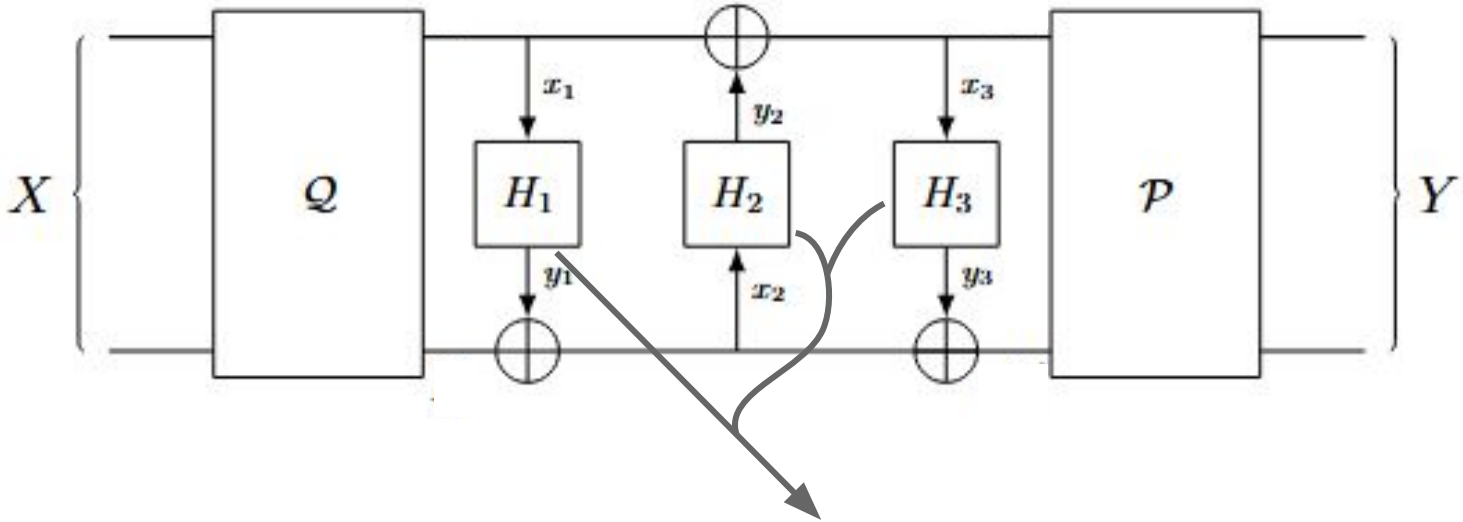


Compressed oracle: record H_1, H_2, H_3 queries

(X, Y) was queried ⇔ ∃ "consistent chain" recorded

$$Q(X) = (x_1, y_1 \oplus x_2) \wedge P(x_3, x_2 \oplus y_3) = Y \wedge x_1 \oplus x_3 = y_2$$

Lifting Query Recording



Compressed oracle: record H_1, H_2, H_3 queries

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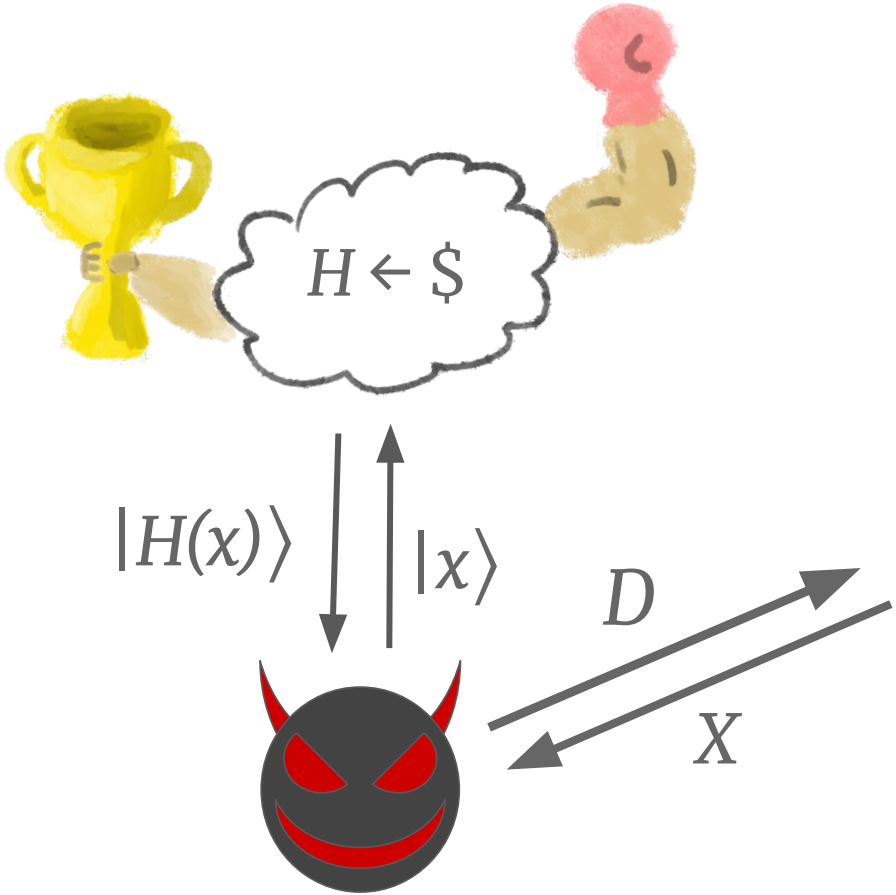
We **recover existing bounds**: double-sided zero search

and **obtain new bounds**: finding k > q input-output pairs in q queries

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Lifting GHHM Reprogramming

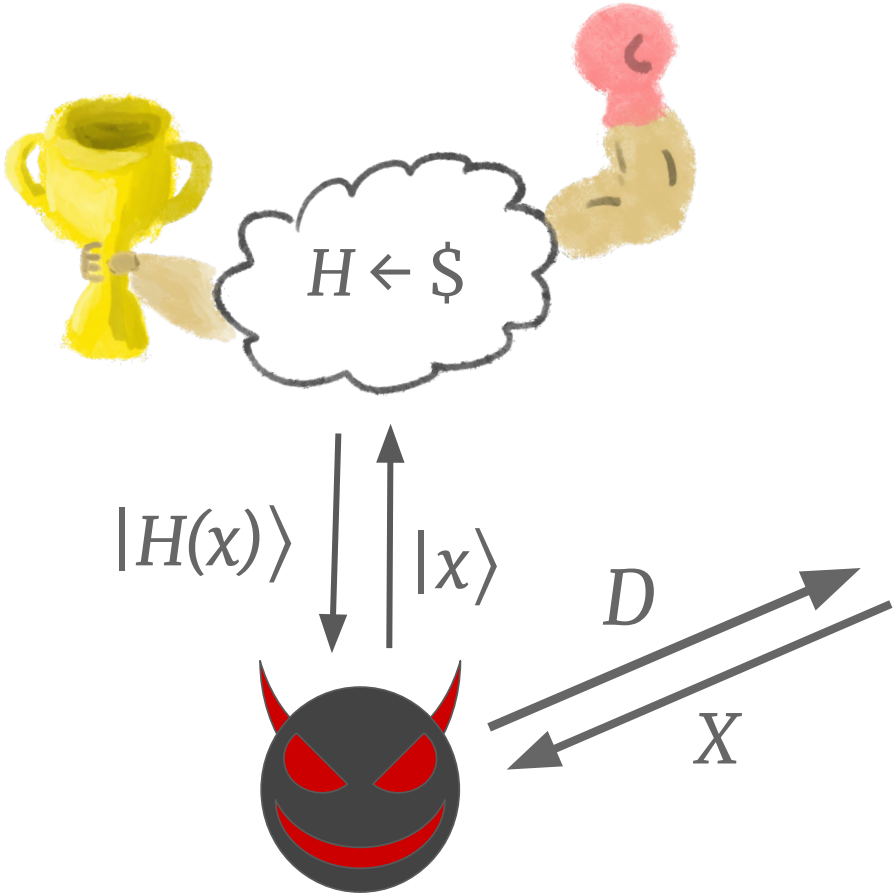


No Prog(D)
1. $X \leftarrow D$

or

Prog(D)
1. $X \leftarrow D$
2. $H(X) := \$$

Lifting GHHM Reprogramming



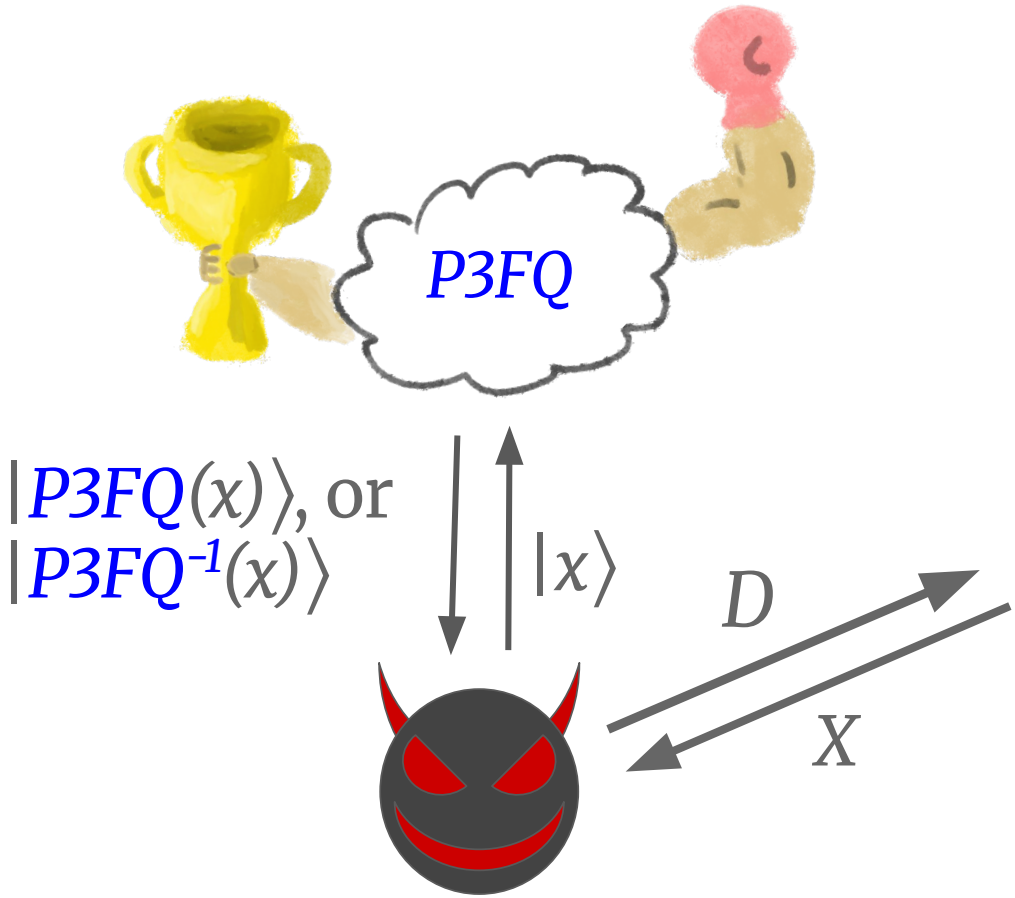
GHHM Theorem.
Indistinguishable if D has high min-entropy.

No Prog(D)
1. $X \leftarrow D$

or ≈

Prog(D)
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Lifting GHHM Reprogramming



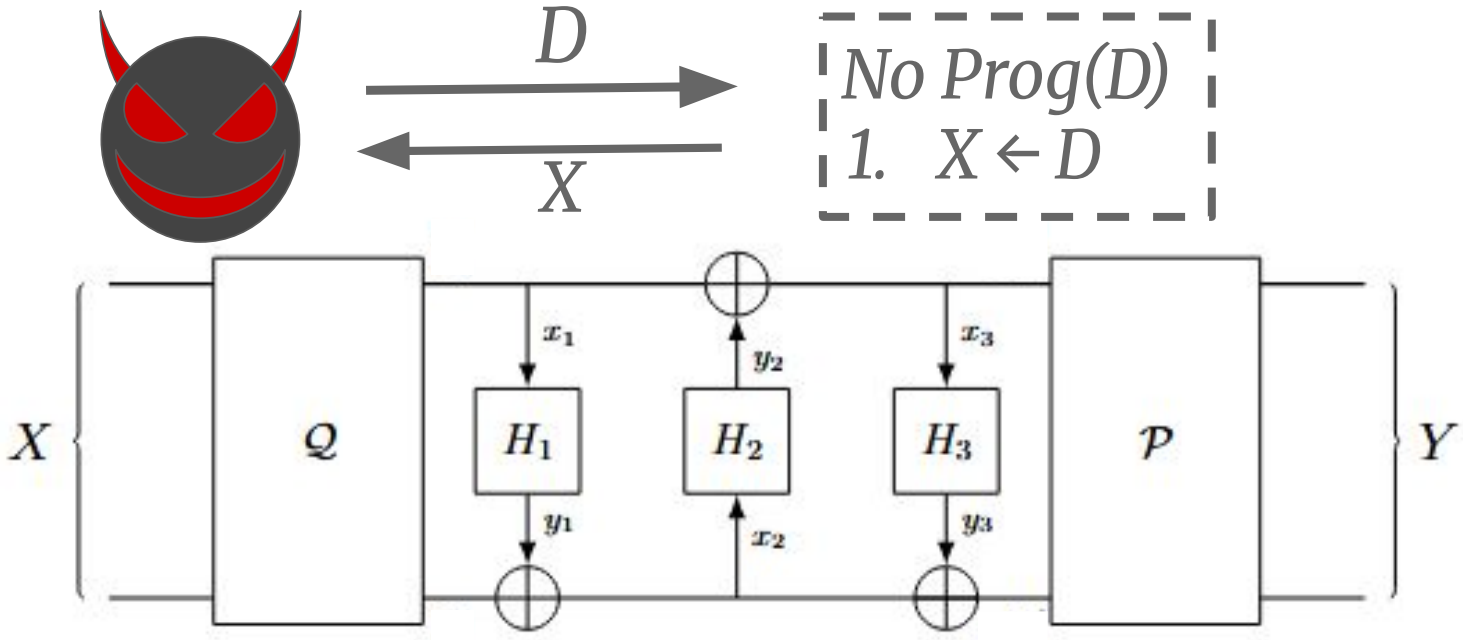
Our lifting.
Indistinguishable if D has high min-entropy.

No Prog(D)
1. $X \leftarrow D$

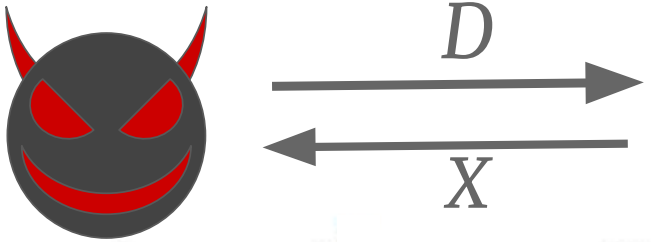
or ≈

Prog(D)
1. $X \leftarrow D$
2. *"P3FQ(X) := $"*

Lifting GHHM Reprogramming



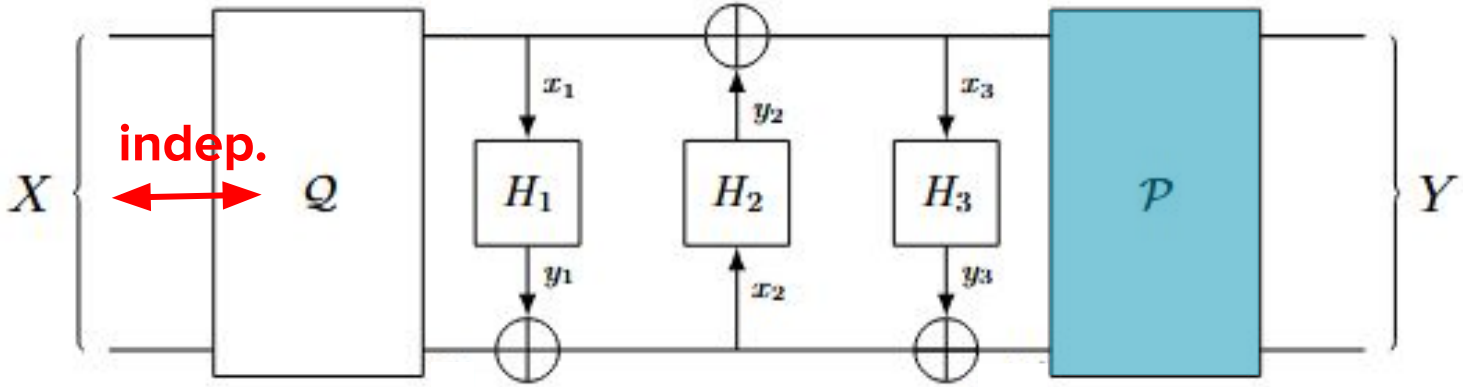
Lifting GHHM Reprogramming



D

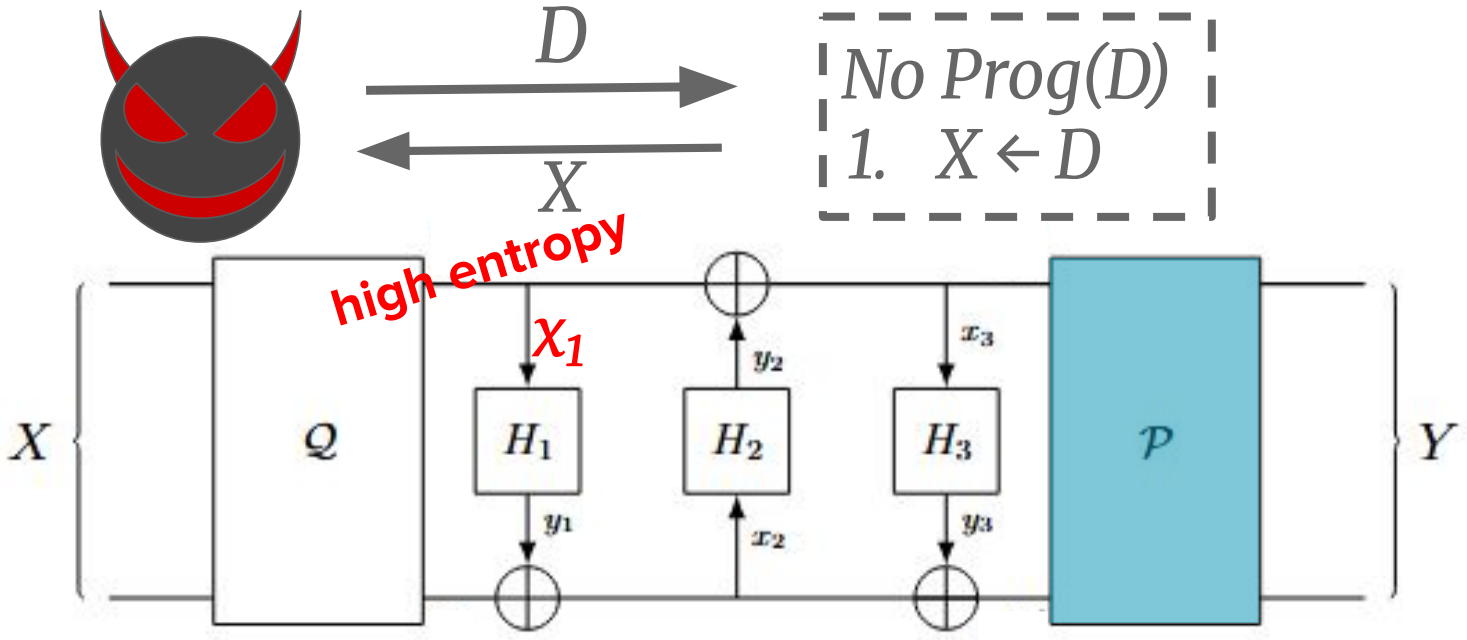
X

$No\ Prog(D)$
$1.\ \ X \leftarrow D$



P is masking Q ⇒ X is indep. of Q

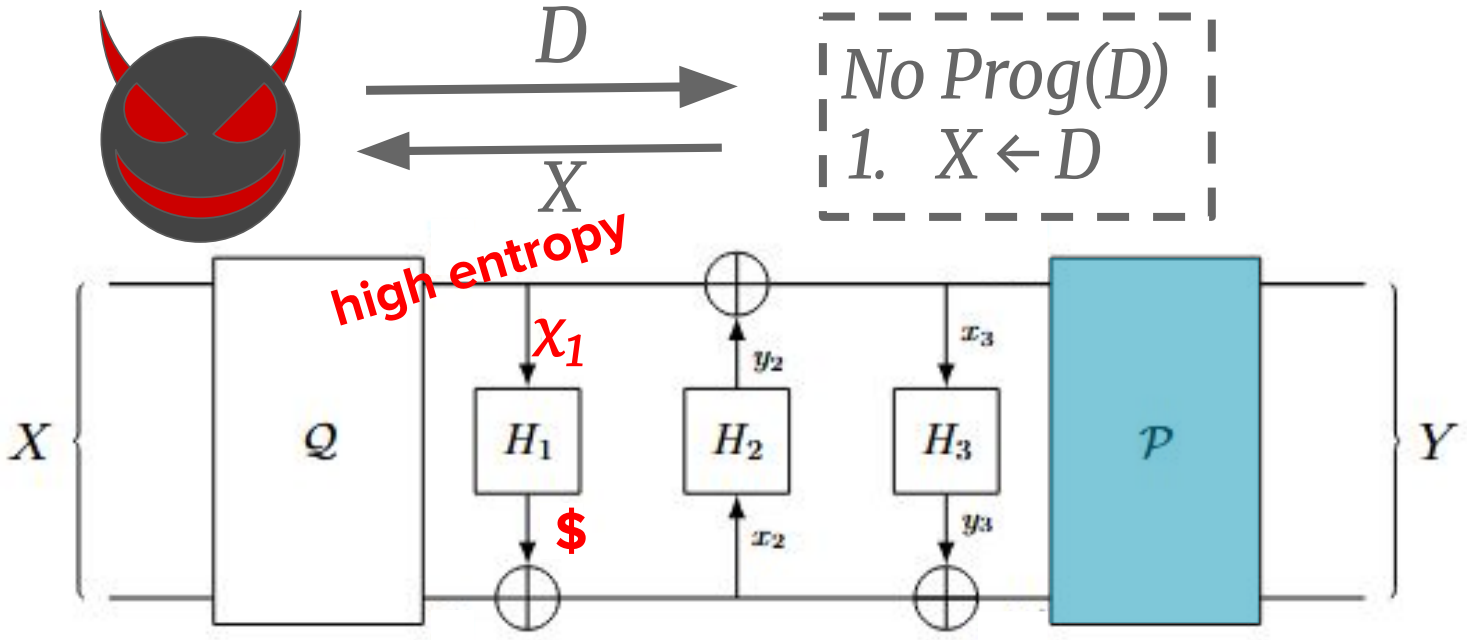
Lifting GHHM Reprogramming



P is masking Q ⇒ X is indep. of Q

⇒ $H\infty($ x_1 := left wire of $Q(X)$ | Q $) \approx H\infty(X)$

Lifting GHHM Reprogramming

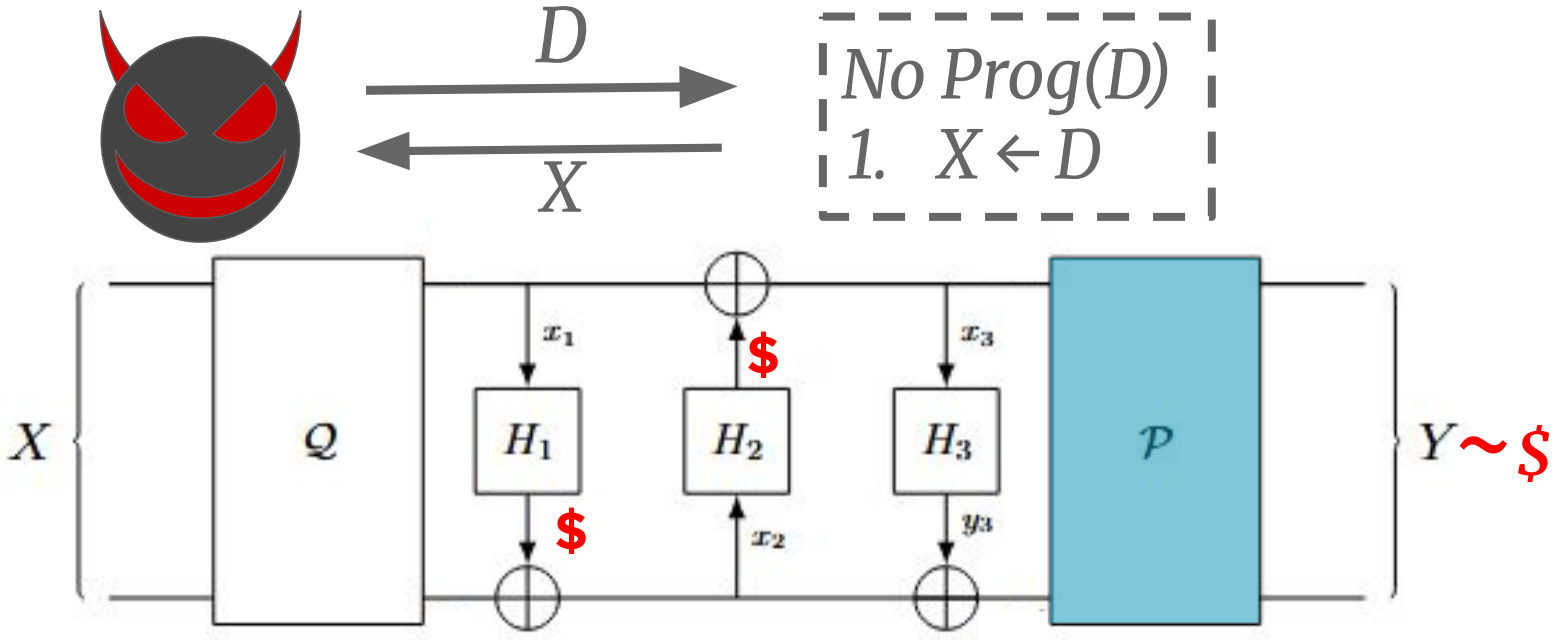


P is masking Q ⇒ X is indep. of Q

⇒ $H\infty($ x_1 := left wire of $Q(X)$ | Q $) \approx H\infty(X)$

⇒ apply GHHM for "$H_1(x_1) := \$$"

Lifting GHHM Reprogramming

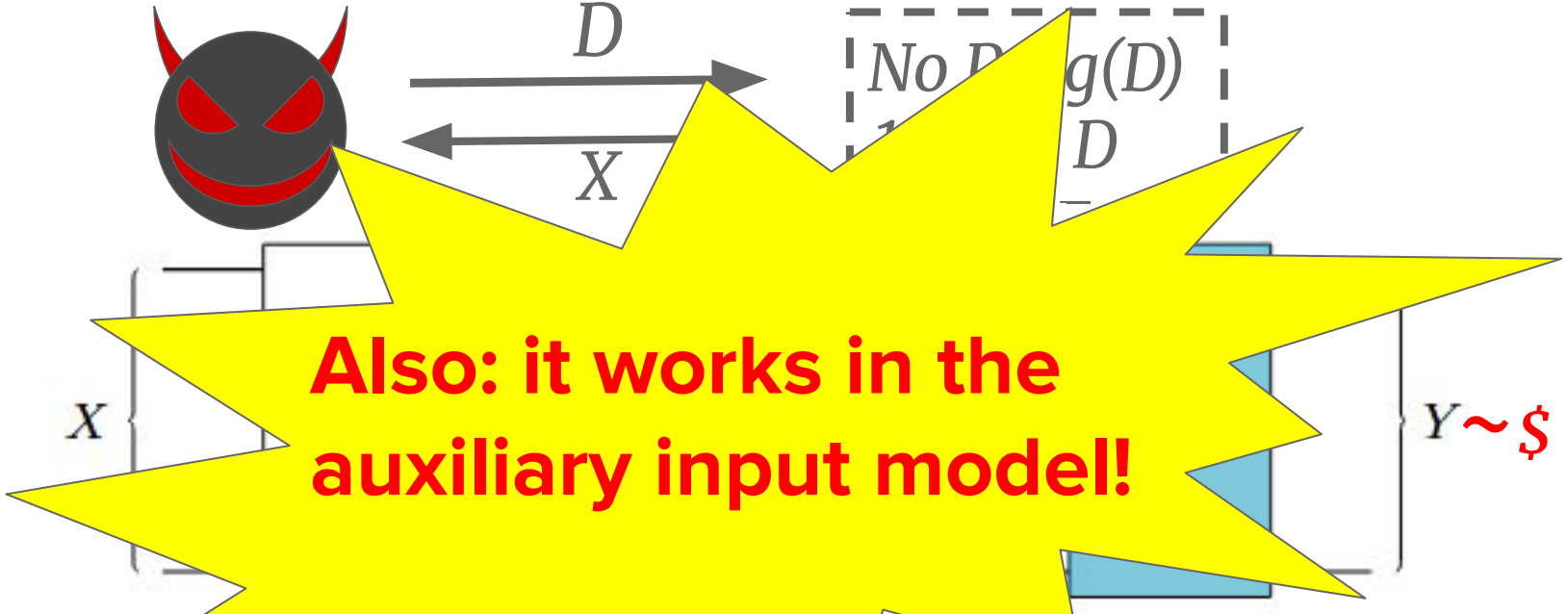


P is masking Q ⇒ X is indep. of Q

⇒ $H\infty(\ x_1 :=$ left wire of $Q(X) \mid Q\ ) \approx H\infty(X)$

⇒ apply GHHM for "$H_1(x_1) :=$ \$" and for H_2

Lifting GHHM Reprogramming



Also: it works in the auxiliary input model!

P is [illegible] of Q

⇒ $H\infty(\ x_1 :=$ [illegible] wire of $Q(X) \mid Q\ ) \approx H\infty(X)$

⇒ apply GHHM for "$H_1(x_1) := \$$" and for H_2

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Lifting Measure-and-Reprogram

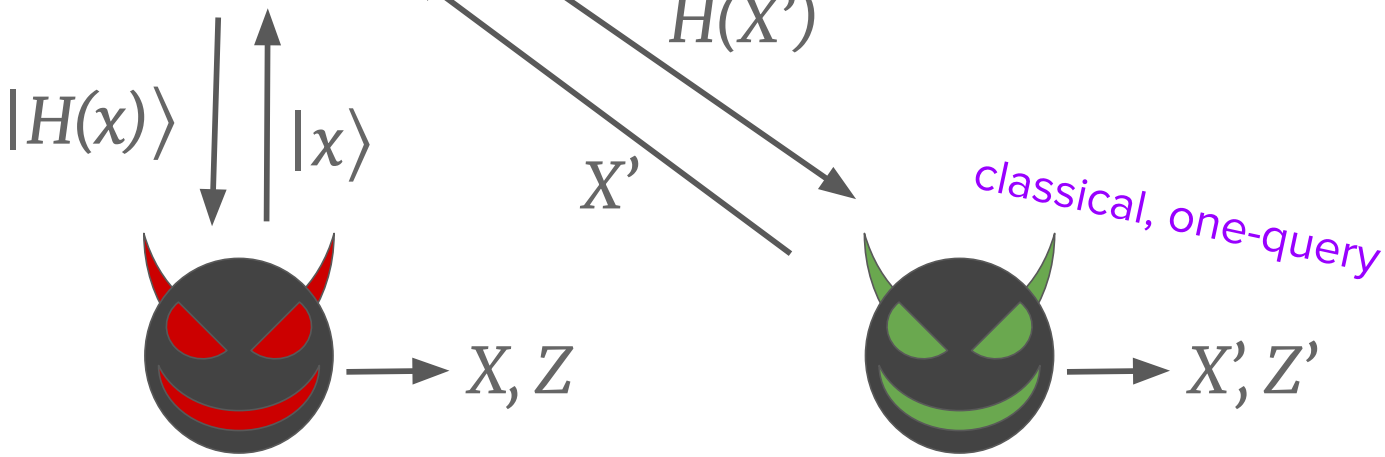


M&R Theorem: ∀  ∃  s.t.

$$Pr[V(X, H(X), Z)]$$

$$\leq O(q^2)\, Pr[V(X, H(X'), Z')]$$

for every predicate $V(\cdot, \cdot, \cdot)$.



Lifting Measure-and-Reprogram



Our M&R: ∀ ∃ **s.t.**

$$Pr[V(X, P3FQ(X), Z)]$$

$$\lesssim O(q^4)\, Pr[V(X, P3FQ(X'), Z')]$$

for "sparse" predicate $V(\cdot, \cdot, \cdot)$.

$|P3FQ(x)\rangle$, or $|P3FQ^{-1}(x)\rangle$

$|x\rangle$

$P3FQ(X')$

X'

classical, one-query

X, Z

X', Z'

Lifting Measure-and-Reprogram



Our M&R: ∀ ∃ **s.t.**

$$Pr[V(X, P3FQ(X), Z)]$$

$$\lesssim O(q^4)\, Pr[V(X, P3FQ(X'), Z')]$$

for "sparse" predicate $V(\cdot,\cdot,\cdot)$.

Intuition: apply M&R to round functions, and the sparseness enforce forward direction.

$|P3FQ(x)\rangle$, or $|P3FQ^{-1}(x)\rangle$

$|x\rangle$

X'

classical, one-query

X, Z

X', Z'

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Applications: Duplex-Sponge Fiat-Shamir (DSFS)

a permutation-based variant of Fiat-Shamir, considered by Signal

A Fiat–Shamir Transformation From Duplex Sponges

Alessandro Chiesa
alessandro.chiesa@epfl.ch
EPFL

Michele Orrù
m@orru.net
CNRS

July 9, 2025

Abstract

We analyze a variant of the Fiat–Shamir transformation based on an ideal permutation. The transformation relies on the popular duplex sponge paradigm, and minimizes the number of calls to the permutation (given the information to absorb/squeeze). This closely models deployed variants of the Fiat–Shamir transformation, and our analysis provides concrete security bounds to guide security parameters in practice. Our results contribute to the ongoing community-wide effort to achieve rigorous, and ultimately formally verified, security proofs for deployed cryptographic proofs. Along the way, we find that indifferentiability (a property proven for many modes of operation, including the duplex sponge) is ill-suited for establishing the knowledge soundness and zero knowledge of a non-interactive argument.

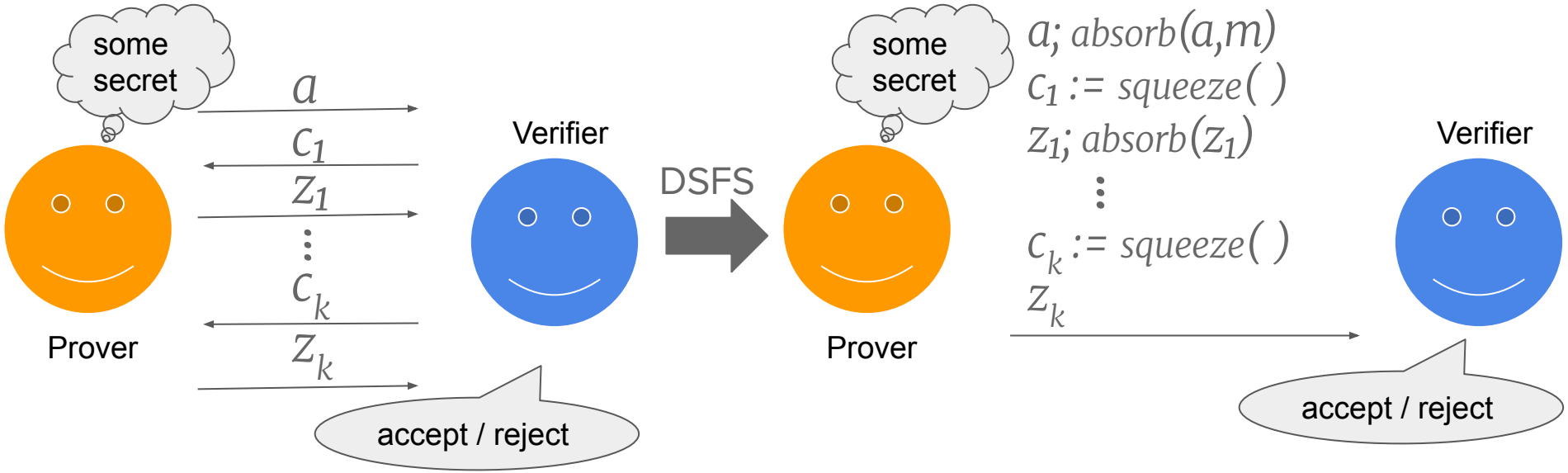
We additionally contribute `spongefish`, an open-source Rust library implementing our Fiat–Shamir transformation. The library is interoperable across multiple cryptographic frameworks, and works with any choice of permutation. The library comes equipped with Keccak and Poseidon permutations, as well as several "codecs" for re-mapping prover and verifier messages to the permutation's domain.

Keywords: Fiat–Shamir transformation; duplex sponge



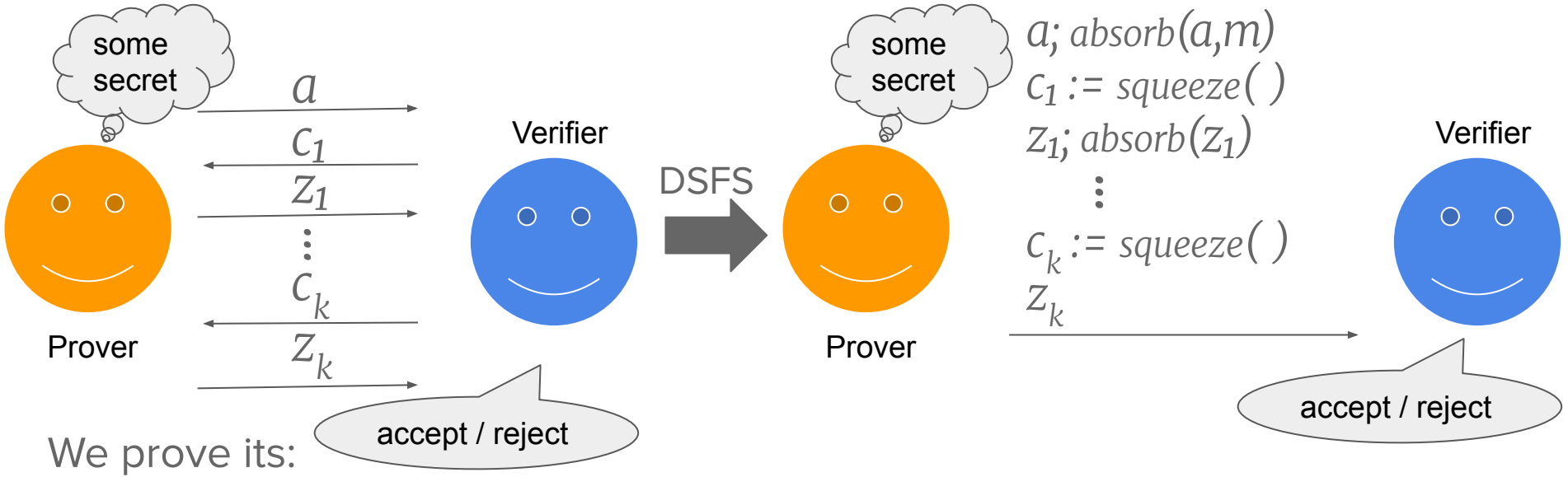
Applications: Duplex-Sponge Fiat-Shamir (DSFS)

a permutation-based variant of Fiat-Shamir, considered by Signal



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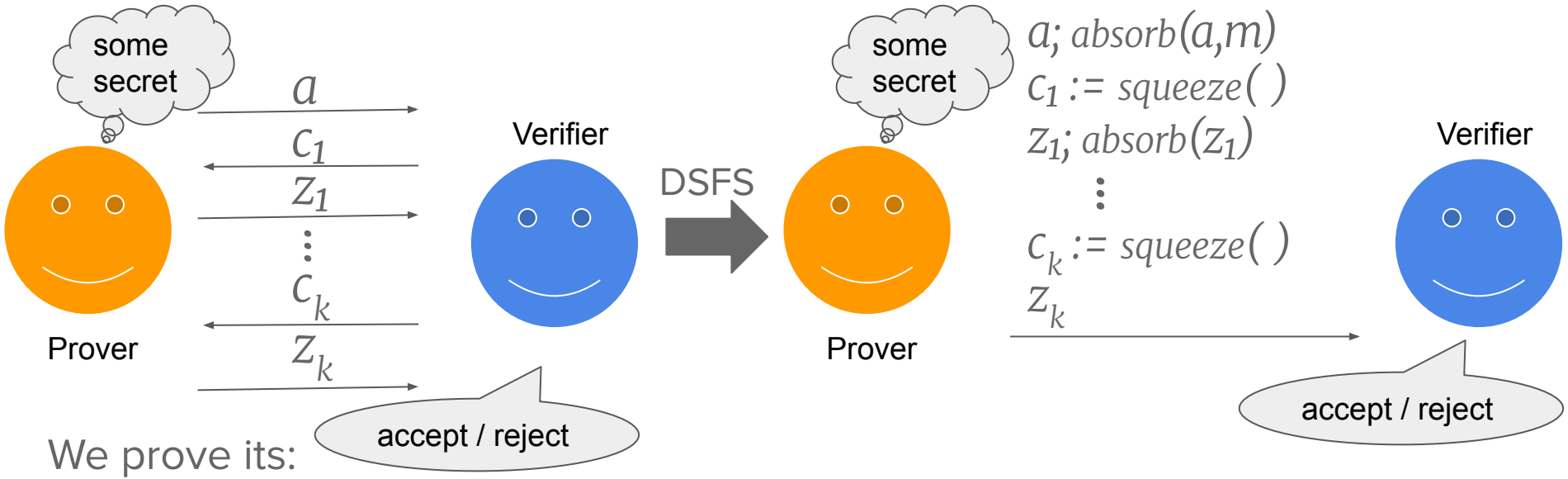
We prove its:

1. post-quantum ZK, and
2. soundness preservation

via lifting 2 different QROM reprogramming techniques to QRPM.

Applications: Duplex-Sponge Fiat-Shamir (DSFS)

a permutation-based variant of Fiat-Shamir, considered by Signal



We prove its:

1. post-quantum ZK (also extends to the auxilliary-input model), and
2. soundness preservation

via lifting 2 different QROM reprogramming techniques to QRPM.

More Applications (WIP)...

Post-Quantum Security of the Even-Mansour Cipher

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with advice!

Abstract. The Even-Mansour cipher is a simple method for constructing a (keyed) pseudorandom permutation E from a public random permutation $P : \{0,1\}^n \to \{0,1\}^n$. It is secure against classical attacks, with optimal attacks requiring q_E queries to E and q_P queries to P such that $q_E \cdot q_P \approx 2^n$. If the attacker is given *quantum* access to both E and P, however, the cipher is completely insecure, with attacks using $q_E, q_P = O(n)$ queries known

More Applications (WIP)...

Post-Quantum Security of the Even-Mansour Ci

Gorjan Alagic[1], Chen Bai[2], Jo
Maje

[1] QuICS, Un
[2] Dept. of Electrical and C
[3] Dept. of Comput
[4] Dept. of Applied Mathematics

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Tight Quantum Time-Space Tradeoffs for Permutation Inversion

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October 15, 2025

simpler hybrid proof!

Abstract

In permutation inversion, we are given a permutation $\pi : [N] \to [N]$, and want to prepare some advice of size S, such that we can efficiently invert any image in time T. This is a fundamental cryptographic problem with profound connections to communication complexity and circuit lower bounds.
lassical setting, a tight $ST = \tilde{\Theta}(N)$ bound has been established since the seminal

More Applications (WIP)...



Thank you for your listening.

ia.cr/2026/146



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